

Galactix Solutions: Code Architecture, AI Tool Usage & Code Review Policy

1. Purpose

This structural policy governs code quality standards, algorithmic integrity, and the guidelines for using automated software engineering utilities and generative Artificial Intelligence tools within Galactix Solutions development lifecycles.

2. Generative AI Tools & LLM Assistance Guidelines

While Galactix Solutions promotes rapid technological innovation, the ingestion of proprietary codebases into public AI frameworks presents significant legal, copyright, and security risks. Engineers must operate strictly under the following boundaries:

1. **Approved Utilities Only:** Software engineers are permitted to use only enterprise-tier AI assistants specifically provisioned by the IT procurement team (e.g., enterprise instances with data-sharing opt-out toggles enabled).
2. **Code Injection Ban:** Never copy-paste raw proprietary source code, algorithmic structural patterns, or production environmental configurations into consumer-grade, public LLM chat interfaces.
3. **Validation Mandate:** Any code snippet or structural design logic generated by an AI assistant must undergo rigorous manual verification and local sandbox validation before being staged for integration. AI-generated code is not exempt from failure liabilities.

Compliance Check: Automated scanners crawl internal repositories nightly. Detection of unapproved third-party structural frameworks or data leakage indicators will trigger an automatic freeze on the offending branch pending a manual security architecture audit.

3. Structural Code Review Conventions

Maintaining code maintainability requires standardized peer-review procedures. No codebase modification can bypass these safeguards to enter production environments:

- **Mandatory Reviewers:** Every Pull Request (PR) must achieve formal, explicit approval from at least **two senior peer engineers** prior to merging into main branches.

- **Test Coverage Thresholds:** Code modifications must contain structured unit and integration tests. Pull requests that lower the overall repository test coverage below a baseline threshold of **85% coverage** will be rejected by automated CI/CD safety rails.
- **Documentation Updates:** Architectural enhancements or new REST endpoint integrations must include corresponding markdown or OpenAPI specification schema revisions inside the same PR.

4. Continuous Integration / Continuous Deployment (CI/CD) Gates

The code staging infrastructure executes linear compliance gates. Failure within static analysis checkers (linting, vulnerability matrices, code styling setups) blocks deployment pipelines automatically. Exceptions require an approved, time-bound structural waiver from the Principal Architect or engineering lead.